Response to Reviewer FBhK

BioCoLoop, submission 5046 | Revision dated 11 November 2026. Prepared for author review; not posted to OpenReview.

Thank you for your careful review and for recognising the common task interface, the protocol detail and the budget-matched allocation result. We have reframed the paper around its central contribution: **collaborative model development**, in which laboratories whose data cannot be pooled jointly configure, train, evaluate and allocate training for one biological predictor using laboratory-local evidence. Section 4.3 now leads with a same-workflow comparison of ten collaborating laboratories versus one laboratory working alone, evaluated on the same held-out test. The requested same-access random and direct search controls are reported in Section 4.7 and Appendix E.1-E.2, where they characterise how proposal policies use the shared evidence. We also explain the development-test divergence (Appendix E.3) and point to the gene-disjoint grouped evaluation in Appendix E.6.

1. "Training data, development data, and total local computation therefore change alongside the search method" (Weakness 1)

We agree, and we now state explicitly that this is what the main comparison measures: what a laboratory gains by developing a model collaboratively instead of alone. Section 4.3 now leads with **the same Qwen history-enabled BioCoLoop workflow**, run by laboratory 0 alone or by all ten laboratories, on the same held-out tests (Appendix D, Table D.1). Collaboration improves every endpoint: DTI AUROC 89.89 to **93.76**, proteomics AP 20.13 to **36.99**, VCC Top-1 20.22 to **31.54**, Norman Top-1 14.15 to **22.25**, and Tahoe Top-1 15.42 to **22.39**. More search does not let a single laboratory close this gap: single-laboratory random, Bayesian and exhaustive-menu search all remain below collaboration on every endpoint (for example, DTI 90.10 versus 93.76).

This collaboration-versus-alone comparison on a common held-out test is the standard evaluation protocol in federated learning, used by Sheller et al. (2020), Chang et al. (2018, JAMIA), Dayan et al. (2021, EXAM), Ogier du Terrail et al. (2022, FLamby), Warnat-Herresthal et al. (2021, Swarm Learning), Saldanha et al. (2022) and, for single-cell data, Wang et al. (2024, scFed). Federated tuning studies likewise compare tuned predictors under a fixed held-out protocol (Holly et al., 2022; Wang et al., 2023, FedHPO-Bench). Section 4.2 now contains an evaluation-protocol paragraph with these citations and clarifies that every method, including BioCoLoop, returns one predictor.

2. Random search without replacement and direct LLM search with the same ten laboratories (Weakness 1; Question 1)

Appendix E.1-E.2 reports exactly this comparison. Random search without replacement, GP-EI, and direct and history-guided Qwen and Luna share the same ten training laboratories, ten development panels, twelve configurations, initialization and checkpoint-selection rule. Each search receives six proposal slots beyond D00, with 100 rounds per fit and a 700-round cap; random search receives the full search budget, not the fixed model's single-fit budget. Enumerating all **462** six-design subsets gives the exact expected outcome of random search with selection based only on development evidence. Failed LLM requests remain charged against the budget, and shared-first-proposal and realized-fit-count controls are also reported.

We present these controls accurately as characterising how proposal policies use shared evidence. The effect of history is task-dependent and conventional search is competitive. The clearest positive case is Norman, where Luna history reaches **23.11** Top-1 versus 22.30 for direct search and 22.66 for random search. DTI favours direct Luna and GP-EI (94.60 versus 94.27 for history), proteomics ties throughout, and on VCC the LLM searches and GP-EI retain 31.50 while random search averages 27.53. Two three-seed follow-ups show the same pattern. On the gene-disjoint cell benchmark (Appendix E.6), structured history reaches 22.888 versus 22.625 for direct and 22.668 for random search, with GP-EI at 23.152. With wider recipe choices (Appendix E.9), Luna structured history reaches DTI AUROC 94.368 versus 94.316 for random search (+0.053 points, conditional 95% interval [0.031, 0.077]), while on proteomics direct search reaches 39.017 AP, matching GP-EI. Section 4.7 now

states this scope explicitly.

Regarding efficiency: we agree that the retrospective target crossings (4 versus 22 proposals on Norman/Luna, 13 versus 5 on Tahoe/Luna) are trajectory diagnostics, not realized savings, because every scheduled proposal was executed. We now describe them that way. The resource result is the separate allocation experiment you highlighted. On Norman, evidence-guided allocation improves Top-1 from 11.48 to **22.32** (+10.84, 95% CI [4.54, 17.73]) at 800 aggregation rounds and 160 development evaluations. New promotion controls (Appendix E.5) show that simpler rules recover much of this gain over uniform allocation, so we present the finding as "early collaborative evidence is useful for allocating training", not as a uniquely superior rule.

3. What explains the development-test divergence in the extended VCC and Tahoe searches? (Weakness 2; Question 2)

Panel construction and weighting. Development and test use disjoint interventions: VCC has 60/20/20 training/development/test interventions and Tahoe 180/60/67. The 20 VCC development interventions form ten panels of two, and the 60 Tahoe compounds form ten panels of six. Because every panel contains the same number of interventions and Top-1 is macro-averaged over interventions, equal weighting of panels equals the pooled intervention-macro score exactly. Equal laboratory weighting therefore does not cause the divergence, and differing query counts do not either.

Audit. We reconstructed all 14 selected checkpoints, re-evaluated their 140 development panels and reproduced all 80 archived held-out prefix scores (Appendix E.3). The scores are correct; the divergence arises from selection on a small development population. For example, Tahoe's +5.00 development points correspond to three additional correct development compounds, while the held-out score falls from 25.37 to 13.43.

Roster sensitivity. We re-scored the same initial and final checkpoints on 20 paired development-roster redraws, without retraining or reselection. Qwen/VCC's original +1.17 development advantage becomes -1.34 ± 7.37 (positive in 6/20 draws), and Tahoe's +5.00 becomes -1.33 ± 4.82 (7/20), whereas Luna/VCC remains positive in 12/20 draws per mode. The large declines therefore coincide with development gains that are sensitive to the choice of distractors. The existing trajectories do not separate repeated-selection effects from biological shift, and we state this in Appendix E.3.

Other grouped partitions. Appendix E.6 provides an additional grouped evaluation: four Replogle/Nadig study-cell contexts with gene-disjoint partitions of 1,208/414/406 training/development/test genes over three seeds. Here every final search exceeds the fitted D00 reference (21.43 Top-1), so the large terminal decreases do not recur; we report the one small interim reversal (seed 62, slot 2: +0.057 development, -0.257 test). Moreover, the main-setting laboratories are compound or intervention groups, not random measurements, and source-defined laboratories with one proteomic study each improve target AP from 34.41 to **37.53** and AUROC from 72.33 to **74.39** through collaboration (Appendix B.2).

4. Summary of manuscript changes

The abstract, introduction and conclusion are reframed around collaborative development and the participation result; Section 4.2 adds the protocol paragraph and clarifies that "direct" is feedback-free search, not a single fit; and Appendix A explains why single-laboratory BioCoLoop behaves like direct search: one laboratory has no cross-laboratory evidence to use. Thank you again for questions that have substantially sharpened the paper.

Response to Reviewer AfJB

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Thank you for your expert review, for pointing us to closely related work, and for your positive note on the matched-budget allocation experiment. We have reframed the paper around its central contribution, **collaborative model development**: laboratories whose data cannot be pooled jointly configure, train, evaluate and allocate training for one biological predictor using laboratory-local evidence. Section 4.3 now leads with a same-workflow comparison in which the identical BioCoLoop workflow is run by one laboratory alone or by ten laboratories together on the same held-out test, and collaboration improves all five endpoints. Related work adds a paragraph that distinguishes collaboration among data-holding laboratories from agent-team collaboration (The Virtual Biotech, AutoScientists) and acknowledges Helmsman as directly related prior work on automated federated-learning system synthesis. Same-access search controls (Section 4.7) are now presented as a characterisation of how proposal policies use the shared evidence.

1. "It is difficult to discern what methodological novelty BioCoLoop introduces" relative to The Virtual Biotech, AutoScientists and Helmsman

All three are discussed in the revised related work. The key distinction is **who collaborates and what they contribute**.

In The Virtual Biotech (Zhang et al., 2026), the collaborators are specialist AI agents that retrieve and integrate biomedical evidence for therapeutic discovery and development decisions. In AutoScientists (Gao, Fang and Zitnik, 2026), self-organizing agent teams share a champion program, an experiment log and failed directions to improve computational models. Both are important precedents for coordinated scientific agents and accumulated experimental history. In both, however, the collaborators are agents working on data they can access; neither addresses the setting in which the training and development data are held by different institutions and cannot be pooled.

In BioCoLoop the collaborators are **data-holding laboratories**. Each laboratory trains candidate configurations on its own data and evaluates them on its own development panels; only model updates and bounded development evidence leave the laboratory. That evidence then determines which configuration is retained, what is proposed next and which candidates receive further training. Model development itself, not only parameter fitting, thus becomes collaborative and data-local.

Helmsman (Li, Funk and Saeed, ICLR 2026) is directly related prior work, and we now say so explicitly. Its collaborating LLM agents plan, generate, and diagnose and revise federated-learning systems through sandboxed federated simulations. BioCoLoop is complementary: rather than synthesizing FL system code, it develops a biological predictor inside a fixed collaborative protocol and uses laboratory-local training and evaluation evidence to configure, select and allocate training for that predictor. The revised related work also cites federated configuration search that predates LLM agents, FedEx (Khodak et al., 2021) and FedNAS (He et al., 2020). We do not claim to be the first agentic or automated FL framework. Our contribution is the collaborative biological model-development setting and interface, together with controlled experiments that separate laboratory participation, proposal history and training allocation.

2. "When the number of laboratories is held at one, BioCoLoop performs essentially identically to direct optimization" (Tables A.1-A.3)

Your reading of Table A.1 is correct, and we now explain why this is the expected result. BioCoLoop's mechanism is the use of cross-laboratory development evidence. **A laboratory working alone has no cross-laboratory evidence**, so its proposals and selections largely coincide with feedback-free direct search: 89.89 versus 89.89 on DTI, 20.13 versus 20.13 on proteomics, and 21.09 for direct versus 20.22 on VCC. Appendix A now contains an "Interpreting the single-laboratory control" paragraph that states this.

The informative comparison is therefore the same workflow with and without collaborators. Holding the Qwen history-enabled workflow fixed and changing only participation from

laboratory 0 to ten laboratories (Appendix D, Table D.1), DTI AUROC rises from 89.89 to **93.76**, proteomics AP from 20.13 to **36.99**, and VCC, Norman and Tahoe Top-1 from 20.22, 14.15 and 15.42 to **31.54, 22.25 and 22.39**. Tables A.2 and A.3 show the same pattern: DTI AP 87.77 to 92.73, proteomics AUROC 61.80 to 73.87, unseen-drug DTI AUROC 88.39 to 91.65 and unseen-protein DTI AUROC 83.26 to 87.63. More extensive local search does not close this gap: single-laboratory random, Bayesian and exhaustive-menu search remain below collaboration on every endpoint (for example, DTI 90.10 versus 93.76).

3. "The main comparison in Table 1 provides BioCoLoop access to ten laboratories, while the existing AI-scientist baselines operate on a single laboratory"

We now state explicitly that this is the intended comparison: what a laboratory gains by developing a model with collaborators rather than alone. Comparing collaborative and single-site training on a common held-out test is the standard evaluation protocol in federated learning (Sheller et al., 2020; Chang et al., 2018; Dayan et al., 2021, EXAM; Ogier du Terrail et al., 2022, FLamby; Warnat-Herresthal et al., 2021; Saldanha et al., 2022; Wang et al., 2024, scFed), including federated hyperparameter tuning (Holly et al., 2022; Wang et al., 2023, FedHPO-Bench). Section 4.2 now adds this protocol paragraph and clarifies that every method returns one predictor and that "direct" is feedback-free search, not a single fit.

We agree that Table 1 by itself changes both the workflow and the data access. This is why Section 4.3 now leads with the same-workflow comparison above, in which only participation changes. The AI-scientist baselines remain as references for what current agents achieve with one laboratory's data.

4. "Across ten comparisons, eight show identical final performance" (Section 4.7)

This observation is accurate, and the revised Section 4.7 reports it plainly. The same-access controls hold the ten laboratories, development panels, configuration library and proposal budget fixed, and they compare history-guided proposals with direct LLM search, random search without replacement and GP-EI (Appendix E.1-E.2). The effect of history is task-dependent and conventional search is competitive. The positive cases are Norman, where Luna history reaches **23.11** Top-1 versus 22.30 for direct and 22.66 for random search; the gene-disjoint cell benchmark (Appendix E.6), where structured history reaches 22.888 versus 22.625 for direct and 22.668 for random search; and the wider-recipe DTI study (Appendix E.9), where Luna structured history reaches 94.368 AUROC versus 94.316 for random search (+0.053 points, conditional 95% interval [0.031, 0.077]). Elsewhere, direct search or GP-EI is equal or better.

We therefore present these controls as characterising how proposal policies use shared evidence, not as the source of the main gain. Section 4.7 now states this separation directly.

5. Allocation within collaborative development

The allocation experiment shows a second development decision that collaborative evidence supports. At 800 aggregation rounds and 160 development evaluations for each policy, evidence-guided allocation improves Norman Top-1 from 11.48 to **22.32** (+10.84, 95% CI [4.54, 17.73]). New promotion controls (Appendix E.5) show that simpler promotion rules recover much of this gain over uniform allocation. We present the result as evidence that early laboratory-local measurements are useful for allocating a fixed training budget, not as a new allocation algorithm.

Thank you again; your comments have made both the positioning and the evaluation considerably clearer.

Response to Reviewer cR5g

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Thank you for your review and for highlighting the importance of distributed training when laboratories cannot share data. We have reframed the paper around its central contribution, **collaborative model development**: laboratories whose data cannot be pooled jointly configure, train, evaluate and allocate training for one biological predictor using laboratory-local evidence. Section 4.3 now leads with a same-workflow comparison in which the identical BioCoLoop workflow is run by one laboratory alone or by ten laboratories together on the same held-out test, and collaboration improves all five endpoints. Random, Bayesian and exhaustive search are reported both for a single laboratory (Appendix D) and at matched ten-laboratory access (Section 4.7, Appendix E.1-E.2). Section 4.2 now clarifies that "direct" is feedback-free configuration search, not a single fit, and that every method returns one predictor.

1. "No comparison to traditional hyperparameter tuning, not even to random search"

Can one laboratory tune its way to the collaborative result? No. Appendix D runs random search, GP-EI Bayesian optimization and exhaustive enumeration of the configuration menu on laboratory 0. All remain below collaboration on every endpoint; for example, DTI AUROC is 90.10 for the conventional controls versus **93.76** for collaboration, and proteomics AP is 20.13 for random search versus **36.99**. Holding the history-enabled BioCoLoop workflow fixed, ten laboratories improve on laboratory 0 alone from 89.89 to 93.76 (DTI), 20.13 to 36.99 (proteomics), 20.22 to 31.54 (VCC), 14.15 to 22.25 (Norman) and 15.42 to 22.39 (Tahoe) (Appendix D, Table D.1).

With the same ten laboratories, how do proposal policies compare? Appendix E.1-E.2 compares random search without replacement (the exact expectation over all 462 six-design subsets), GP-EI, and direct and history-guided LLM search with identical laboratories, development panels, configuration library and proposal budget; failed LLM requests remain charged. Here the effect of history is task-dependent and conventional search is competitive, which we report plainly. Positive cases include Norman, where Luna history reaches **23.11** Top-1 versus 22.30 for direct and 22.66 for random search, and the gene-disjoint cell benchmark (Appendix E.6), where structured history reaches 22.888 versus 22.625 for direct and 22.668 for random search. On DTI, GP-EI reaches 94.60 and random search 94.45, against 94.27 for history. These controls characterise how proposal policies use shared evidence; the main contribution is collaborative development itself.

2. "The laboratories are not real independent entities with a domain shift and batch effects, but random splits of a single dataset"

First, the main-setting laboratories are **not random splits of measurements**. They are grouped by biological unit: DTI laboratories are defined by canonical compounds, proteomic laboratories by compound groups, and cell laboratories hold disjoint sets of training interventions (one mean profile per intervention), so no compound or intervention is shared between laboratories. Development and test interventions are disjoint from training. Simulating clients by partitioning one dataset is the standard protocol in federated-learning research (McMahan et al., 2017; Caldas et al., 2018, LEAF; He et al., 2020, FedML; Beutel et al., 2020, Flower), and Section 4.2 now states this with citations.

Second, we do test the harder case you describe. In **source-defined** experiments, each laboratory is a different released study with its own experimental context, measurement protocol and response objective (Appendix B.2). With one proteomic study per laboratory (PTPC, Lin, Ruprecht and decryptE, combined through a shared representation with source-specific heads), collaboration raises target AP from 34.41 to **37.53** and AUROC from 72.33 to **74.39**. For DTI, adding Davis or Human raises target AUROC from 85.47 to 87.86 and 87.95. For cells, adding the Jiang study raises VCC Top-1 from 29.49 to 33.07. We report every source, including those that do not help, because compatibility between studies matters. Execution is simulated on shared compute, and we do not claim formal privacy guarantees.

3. "The authors' own pipeline, when training a single model ('direct'), performs worse than a fixed model"

"Direct" is not a single model fit. It is configuration search without feedback from earlier evaluations: several candidate configurations are trained, and one is retained according to development score and loss. "Fixed" is one predefined configuration. Both return a single predictor. Section 4.2 now states this explicitly.

In the single-laboratory setting, direct search selects using only laboratory 0's development data, which is small. A configuration that wins on that set need not win on the test; for example, single-laboratory direct proteomics scores 20.13 AP versus 25.05 for the fixed model. Collaboration addresses exactly this limitation: with ten laboratories' training and development evidence, BioCoLoop reaches **36.99** AP on the same test. Collaboration also exceeds the single-laboratory fixed model on DTI, proteomics, VCC and Tahoe; on Norman, the fixed configuration (25.56) remains a strong reference, and we report this. Appendix E.3 further audits development selection in the ten-laboratory extended searches and shows that small development gains there can depend on the candidate roster.

4. "Other baselines also use 1 model, while BioCoLoop uses 10 labs"

The ten laboratories are data holders, not an ensemble: BioCoLoop, like every baseline, returns **one predictor**, evaluated on the same held-out test. The comparison asks what a laboratory gains by developing that predictor with collaborators instead of alone. This is the standard evaluation in federated learning (Sheller et al., 2020; Chang et al., 2018; Dayan et al., 2021; Ogier du Terrail et al., 2022, FLamby; Warnat-Herresthal et al., 2021; Saldanha et al., 2022; Wang et al., 2024, scFed), including federated tuning (Holly et al., 2022; Wang et al., 2023, FedHPO-Bench). Section 4.3 now leads with the same BioCoLoop workflow run by one versus ten laboratories (Section 1 above), so that only participation changes.

5. "The main approach appears to be asking an LLM to turn 4 knobs"

The contribution is not a new optimizer over four hyperparameters; it is **making model development itself collaborative** across laboratories that cannot pool data. BioCoLoop provides a common interface through which laboratories train candidate models locally, evaluate them on their own development data, and share model updates and bounded development evidence. That evidence then drives configuration selection, subsequent proposals and the allocation of further training, across three biological task families. The original four-factor library is a deliberately controlled instantiation that makes the participation, history and allocation comparisons interpretable. Appendix E.9 widens it to 36 recipes that also vary the prediction head, the training objective and the learning-rate schedule. There, direct search reaches 39.017 proteomics AP versus 37.109 for the fitted default, and DTI search reaches 94.368 AUROC versus 94.271.

Allocation is a second collaborative development decision. At a fixed 800 aggregation rounds and 160 development evaluations, evidence-guided allocation raises Norman Top-1 from 11.48 to **22.32** (+10.84, 95% CI [4.54, 17.73]); simpler promotion rules recover much of this gain over uniform allocation (Appendix E.5). The revised related work distinguishes this laboratory-level collaboration from agent-team systems and from Helmsman's automated synthesis of FL systems, which we acknowledge as directly related prior work.

Thank you again; these comments have clarified both the contribution and the evaluation protocol.